

# DIGITAL FORENSIC INVESTIGATION REPORT

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BitTorrent Activity Investigation on a Windows 10 Enterprise System

## CASE REFERENCE

BTD-2026-001

## EXAMINER

Paulo Vaz

## REPORT DATE

June 28, 2026

## REPORT VERSION

1.0 — Final

## CLASSIFICATION

Academic Portfolio

## EVIDENCE IMAGE (MD5)

8e01029edeadbffffb36fe8516afb54df

## STANDARDS REFERENCED

ISO/IEC 27037, 27041, 27042, 27043 · NIST SP 800-86

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*This report documents a digital forensic investigation conducted in a controlled laboratory environment for academic and portfolio purposes. All findings, methodologies, and conclusions are the work of the examiner and reflect standard forensic practice.*

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## 1. Identification

|                            |                                                    |
|----------------------------|----------------------------------------------------|
| Case Reference             | BTD-2026-001                                       |
| Examiner                   | Paulo Vaz                                          |
| Requesting Party           | Academic Portfolio — Self-initiated                |
| Examination Date (Disk)    | June 23–24, 2026                                   |
| Examination Date (Network) | June 28, 2026                                      |
| Report Date                | June 28, 2026                                      |
| Report Version             | 1.0 — Final                                        |
| Classification             | Academic Portfolio                                 |
| Standards Referenced       | ISO/IEC 27037, 27041, 27042, 27043; NIST SP 800-86 |

## 2. Executive Summary

This report documents the findings of a digital forensic investigation conducted on a Windows 10 Enterprise LTSC virtual machine identified as **windows-10-dfir-target**. The investigation was carried out in a controlled laboratory environment as part of an academic portfolio project.

The examination sought to determine whether BitTorrent software was installed and used on the target system, and to reconstruct the complete timeline of activity — from software acquisition to torrent download — through independent analysis of disk artifacts and network traffic. The investigation spanned six phases: environment setup, evidence generation, forensic acquisition, disk analysis, network analysis, and consolidated reporting. Two independent forensic tools were employed — Autopsy 4.23.1 and The Sleuth Kit 4.14.0 — with cross-validation between tools used as a methodological control.

### Principal findings:

|                                                    |                                                                                                                                                                                                                                                     |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>qBittorrent 5.2.2 installation confirmed</b>    | Downloaded from sourceforge.net on 2026-06-19 at 10:29 BRT and first executed at 10:33:04 BRT, per Prefetch artifacts and browser download history.                                                                                                 |
| <b>Torrent download confirmed</b>                  | The file debian-13.5.0-amd64-netinst.iso (755 MiB) was downloaded via BitTorrent on 2026-06-22 between 12:29:33 and 12:30:10 BRT — 37 seconds — confirmed independently by application logs, NTFS timestamps, browser history, and network traffic. |
| <b>Network evidence corroborates disk evidence</b> | The SHA-1 info-hash 58846860f0a766f8a42b0bb214d8c713fdf1b167, recovered from the .torrent filename on disk, was independently identified in BitTorrent handshake payloads captured in the network trace during the exact download window.           |
| <b>No evidence of tampering</b>                    | NTFS timestamp analysis (\$STANDARD_INFORMATION = \$FILE_NAME) confirmed no timestamping on the .torrent file.                                                                                                                                      |
| <b>64 peer connections established</b>             | The target established 64 TCP connections to external peers during the download window, consistent with normal BitTorrent swarm behaviour.                                                                                                          |

## 3. Objective

The investigation was conducted with the following objectives:

1. Confirm whether qBittorrent was installed on the target system and establish the installation timeline.
2. Identify and recover all disk artifacts related to BitTorrent activity: configuration files, application logs, browser history, filesystem metadata, and registry entries.
3. Analyze network traffic to independently corroborate disk findings through protocol-level evidence.
4. Reconstruct a unified, chronologically consistent timeline spanning disk and network evidence.
5. Assess artifact integrity, including anti-tampering verification of NTFS timestamps.
6. Produce a formal forensic report documenting all findings, methodology, limitations, and procedural deviations.

## 4. Methodology

The investigation followed a structured six-phase methodology aligned with ISO/IEC 27037 (evidence identification and collection), 27041 (assurance), 27042 (analysis and interpretation), 27043 (incident investigation), and NIST SP 800-86 (integrating forensic techniques into incident response).

|                                        |                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase 01 — Environment Setup</b>    | Two-VM laboratory in VirtualBox 7.2.8: a Windows 10 Enterprise LTSC target and a Kali Linux 2026.1 investigator VM, connected via host-only network.                                                                                                                                                                                       |
| <b>Phase 02 — Evidence Generation</b>  | Installation of qBittorrent 5.2.2 and Wireshark 4.6.6. Download of debian-13.5.0-amd64-netinst.iso via BitTorrent with simultaneous Wireshark capture (585,916 packets).                                                                                                                                                                   |
| <b>Phase 03 — Forensic Acquisition</b> | VDI converted to raw via qemu-img, then E01 acquisition via ewfacquire. The initial acquisition was invalidated due to VirtualBox differential snapshot architecture (base VDI frozen at a pre-evidence state). Remediation and the final image (MD5 8e01029edeadebfff36fe8516afb54df, 9 segments) are documented in the chain of custody. |
| <b>Phase 04 — Disk Analysis</b>        | Analysis with Autopsy 4.23.1 and TSK 4.14.0. Autopsy ingest was interrupted twice; TSK CLI served as an independent validation method for Registry and .torrent metadata. Dual-tool validation strengthens the evidentiary value of the findings.                                                                                          |
| <b>Phase 05 — Network Analysis</b>     | Analysis of the pcap with Wireshark 4.6.6, performed on the Windows target VM (Wireshark was not configured on Kali). This procedural deviation is documented with a full risk assessment in the chain of custody.                                                                                                                         |
| <b>Phase 06 — Consolidated Report</b>  | Synthesis of Phase 04 and Phase 05 findings into a unified timeline and this report.                                                                                                                                                                                                                                                       |

## 5. Case Description

The target system is a Windows 10 Enterprise LTSC 21H2 virtual machine (build 10.0.19044.1288) with 8 GB RAM, 4 virtual CPUs, and a 60 GB dynamic VDI disk. The OS user account is named **Paulo** (SID S-1-5-21-2003726914-3947868171-508885183-1001). The system had two network adapters: NAT (10.0.2.15) and host-only (192.168.56.101).

The scenario involves installation and use of qBittorrent 5.2.2 to download the Debian GNU/Linux 13.5.0 network installer ISO (755 MiB) from the public BitTorrent swarm. Activity occurred across two sessions: a software installation session on June 19, 2026, and a download session on June 22, 2026.

The investigation was initiated as an academic exercise demonstrating end-to-end DFIR methodology — from evidence generation and forensic acquisition through disk and network analysis to formal reporting — using real forensic tools and industry-standard procedures.

## 6. Evidence and Chain of Custody

### 6.1 Primary Forensic Image

|                         |                                                                 |
|-------------------------|-----------------------------------------------------------------|
| Description             | Windows 10 Enterprise LTSC 21H2 virtual machine disk            |
| Source                  | windows-10-dfir-target.vdi (VirtualBox dynamic disk)            |
| Image format            | Expert Witness Format (E01), 9 segments (E01–E09)               |
| Compressed size         | ~13.2 GB                                                        |
| Logical size            | 60 GiB (64,424,509,440 bytes)                                   |
| Acquisition tool        | ewfacquire 20140816                                             |
| Acquisition date        | June 25, 2026                                                   |
| MD5 (stored / verified) | 8e01029edeaddbfff36fe8516afb54df (match)                        |
| Raw SHA-256             | dff2ba411365bbf36cf15306d036f2d24a0ff9bdc895e677ab6f8abab3b4e4e |
| Partition offset (TSK)  | 104448 (vol3, NTFS)                                             |
| Status                  | Active — authoritative image for all analysis                   |

### 6.2 Network Capture

|                 |                                                 |
|-----------------|-------------------------------------------------|
| File            | qbittorrent-debian.pcapng                       |
| Capture tool    | Wireshark 4.6.6                                 |
| Total packets   | 585,916                                         |
| Capture window  | 2026-06-22 12:27 – 12:32 BRT                    |
| Download window | 2026-06-22 12:29:33 – 12:30:10 BRT (37 seconds) |
| Status          | Secondary evidence — corroborates disk findings |

### 6.3 Chain of Custody Notes

An initial acquisition (MD5 41f3b630078af93662a45f37d3d6ee9b) was **invalidated** after Phase 04 analysis revealed it captured the VDI base disk frozen at the clean-install snapshot, prior to any evidence generation. Root cause: VirtualBox differential snapshot architecture — all evidence resided in snapshot differential VDIs, not the base VDI.

Before remediation, the complete VM directory, including all snapshot differentials, was backed up to two independent devices (external HD, NTFS; USB drive, exFAT). The original snapshot chain is

preserved and verifiable.

**Procedural deviation.** Network capture analysis (Phase 05) was performed on the evidence-generating machine rather than the investigator workstation, because Wireshark was not configured on Kali Linux. This deviation is explicitly acknowledged and would not be acceptable in a production forensic or legal context. A full risk assessment is recorded in the chain of custody.

## 7. Forensic Timeline

The following unified timeline was reconstructed from four independent evidence sources: browser download history, Windows Prefetch, the qBittorrent application log, NTFS timestamps, and network traffic analysis.

| Timestamp (BRT)        | Event                                                           | Source                        |
|------------------------|-----------------------------------------------------------------|-------------------------------|
| 2026-06-19 10:27       | Wireshark 4.6.6 downloaded from wireshark.org                   | Web Downloads                 |
| 2026-06-19 10:29       | qBittorrent 5.2.2 downloaded from sourceforge.net               | Web Downloads                 |
| 2026-06-19 10:33:04    | qBittorrent installed and first launched (PID 3716, port 29823) | Prefetch, log                 |
| 2026-06-19 10:33:05–17 | DHT/PeX/encryption enabled; external IP 187.106.172.158         | Application log               |
| 2026-06-19 10:33:18    | Session 1 ended — no torrent activity                           | Application log               |
| 2026-06-22 12:27       | Wireshark capture started                                       | pcap                          |
| 2026-06-22 12:28:15    | qBittorrent launched (PID 8060, port 29823)                     | Application log               |
| 2026-06-22 12:29:33    | Torrent added: debian-13.5.0-amd64-netinst.iso                  | Log, NTFS \$SI, Web Downloads |
| 2026-06-22 12:29:33    | .torrent created in BT_backup (inode 156645); \$SI=\$FN         | TSK istat                     |
| 2026-06-22 12:29:33    | HTTP GET /announce to 130.239.18.158:6969 (event=started)       | pcap frame 14610              |
| 2026-06-22 12:29:33    | Tracker 200 OK — 100 peers returned                             | pcap frame 14680              |
| 2026-06-22 12:29:33    | DHT Get_peers queries begin for target info-hash                | pcap frames 14474+            |
| 2026-06-22 12:29:35    | Web seed error: cdimage.debian.org 404 (web seed, not tracker)  | Application log               |
| 2026-06-22 12:29:35+   | BitTorrent handshakes with peers — info-hash in payload         | pcap frame 16107+             |
| 2026-06-22 12:30:10    | Download complete: 755 MiB in 37 seconds                        | Application log               |
| 2026-06-22 12:32       | Wireshark capture stopped                                       | pcap                          |

## 8. Technical Analysis

### 8.1. Disk Analysis (Phase 04)

Disk analysis was performed against the E01 image using Autopsy 4.23.1 and TSK 4.14.0, with findings confirmed via both tools where applicable. Autopsy ingest was interrupted twice; TSK CLI served as the primary validation method for Registry and .torrent metadata extraction.

#### 8.1.1 Partition Layout

| ID | Volume | Start Sector | Length (Sectors) | Type           | Flags       |
|----|--------|--------------|------------------|----------------|-------------|
| 1  | vol1   | 0            | 2,048            | Unallocated    | Unallocated |
| 2  | vol2   | 2,048        | 102,400          | NTFS/exFAT     | Allocated   |
| 3  | vol3   | 104,448      | 124,656,856      | NTFS/exFAT     | Allocated   |
| 4  | vol4   | 124,761,304  | 808              | Unallocated    | Unallocated |
| 5  | vol5   | 124,762,112  | 1,062,912        | Unknown (0x27) | Allocated   |
| 6  | vol6   | 125,825,024  | 4,096            | Unallocated    | Unallocated |

**Note.** vol3 (start sector 104448) is the primary NTFS Windows partition containing all user data and evidence artifacts. This offset was confirmed via TSK prior to Autopsy analysis.

#### 8.1.2 .torrent File

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Inode                   | 156645 (sequence 3)                                   |
| Short name (8.3)        | 588468~1.TOR                                          |
| Full name               | 58846860f0a766f8a42b0bb214d8c713fdf1b167.torrent      |
| Full path               | Users/Paulo/AppData/Local/qBittorrent/BT_backup/      |
| Parent MFT Entry        | 96082                                                 |
| Allocated / Actual size | 98,304 / 60,608 bytes                                 |
| Owner SID               | S-1-5-21-2003726914-3947868171-508885183-1001 (Paulo) |
| Source                  | TSK fls -r -p -o 104448                               |

The filename is the SHA-1 info-hash of the torrent — the unique identifier used by the BitTorrent protocol to identify this torrent in the DHT network and among peers. This value was independently recovered from network traffic in Phase 05.

Torrent metadata (bencoding):

|               |                                                                    |
|---------------|--------------------------------------------------------------------|
| comment       | Debian CD from cdimage.debian.org                                  |
| created by    | mktorrent 1.1                                                      |
| creation date | 1778933537 (Unix) = 2026-05-16 — Debian project, not download date |
| name          | debian-13.5.0-amd64-netinst.iso                                    |
| length        | 791,674,880 bytes (755 MiB)                                        |
| piece length  | 262,144 bytes (256 KB)                                             |



|        |                     |
|--------|---------------------|
| pieces | 60,400 SHA-1 hashes |
|--------|---------------------|

### 8.1.3 NTFS Timestamps and Anti-Tampering

NTFS maintains two independent timestamp sets per file: \$STANDARD\_INFORMATION (\$SI), modifiable via Windows APIs, and \$FILE\_NAME (\$FN), written only by the NTFS kernel driver. Timestamping modifies \$SI while leaving \$FN unchanged, producing a detectable discrepancy. Identical \$SI and \$FN values indicate natural file creation.

| Attribute              | Created  | Modified | MFT Mod. | Accessed |
|------------------------|----------|----------|----------|----------|
| \$STANDARD_INFORMATION | 12:29:33 | 12:29:33 | 12:29:33 | 12:29:33 |
| \$FILE_NAME            | 12:29:33 | 12:29:33 | 12:29:33 | 12:29:33 |

**Result.** \$STANDARD\_INFORMATION = \$FILE\_NAME across all fields (2026-06-22, BRT). No timestamping detected. The timestamp 12:29:33 BRT is corroborated by three independent sources: the application log, NTFS \$SI, and browser download history.

### 8.1.4 Run Programs (Prefetch)

| Prefetch File                     | Program         | Path                       | First Run           | Count |
|-----------------------------------|-----------------|----------------------------|---------------------|-------|
| QBITTORRENT.EXE-17EBDC32.pf       | QBITTORRENT.EXE | /PROGRAM FILES/QBITTORRENT | 2026-06-19 10:33:04 | 3     |
| QBITTORRENT_5.2.2_X64_SETUP.E-... | SETUP.E         | /USERS/PAULO/DOWNLOADS     | 2026-06-19 10:56    | —     |

A run count of 3 for QBITTORRENT.EXE is consistent with the two sessions in the application log plus one additional invocation (likely the installer launching qBittorrent post-installation).

### 8.1.5 Web Downloads

| File                            | Domain             | Timestamp   | Note          |
|---------------------------------|--------------------|-------------|---------------|
| Wireshark-4.6.6-x64.exe         | wireshark.org      | 06-19 10:27 | —             |
| qbittorrent_5.2.2_x64_setup.exe | sourceforge.net    | 06-19 10:29 | 3 CDN mirrors |
| debian-13.5.0-amd64-netinst.iso | cdimage.debian.org | 06-22 12:29 | —             |
| ...iso.Zone.Identifier          | about:internet     | 06-22 12:29 | MotW, Zone 3  |

The Zone.Identifier Alternate Data Stream proves the ISO was downloaded directly from the internet, not copied from a local device. It remains attached to the file even if browser history is deleted.

### 8.1.6 Recent Documents (.lnk)

Windows creates .lnk files in %APPDATA%\Microsoft\Windows\Recent\ when a user opens a file via Explorer or a file dialog — direct evidence of manual user interaction.

| File                      | Target Path                                    | Accessed    |
|---------------------------|------------------------------------------------|-------------|
| debian-...netinst.iso.lnk | C:\Users\Paulo\Downloads\debian-...netinst.iso | 06-22 12:29 |
| debian-...iso.torrent.lnk | C:\Users\Paulo\Downloads\debian-...iso.torrent | —           |

### 8.1.7 Configuration (qBittorrent.ini)

Path: Users/Paulo/AppData/Roaming/qBittorrent/qBittorrent.ini

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| SessionPort      | 29823 — P2P listening port (authoritative for network filter) |
| SessionSSLPort   | 62269                                                         |
| SavePathHistory  | C:\Users\Paulo\Downloads                                      |
| MigrationVersion | 8 (standard qBittorrent 5.x installation)                     |

### 8.1.8 Application Log (qbittorrent.log)

Path: Users/Paulo/AppData/Local/qBittorrent/logs/qbittorrent.log

#### Session 1 — 2026-06-19:

10:33:04 qBittorrent v5.2.2 started. Process ID: 3716  
10:33:05 Listening on TCP/29823, UDP/29823  
10:33:06 DHT: ENABLED | PeX: ENABLED | Encryption: ENABLED  
10:33:06 Binding to IPs: 10.0.2.15, 192.168.56.101, 127.0.0.1  
10:33:17 External IP: 187.106.172.158  
10:33:18 Session shutdown — no torrent activity

#### Session 2 — 2026-06-22:

12:28:15 qBittorrent v5.2.2 started. Process ID: 8060  
12:28:15 Listening on TCP/29823, UDP/29823  
12:29:33 New torrent added: debian-13.5.0-amd64-netinst.iso  
12:29:35 Error from web seed: cdimage.debian.org — 404 Not Found  
12:30:10 Download complete: debian-13.5.0-amd64-netinst.iso

Download duration: 37 seconds for 755 MiB. The 404 at 12:29:35 refers to an HTTP web seed URL, not the BitTorrent tracker; tracker communication succeeded independently (Phase 05).

### 8.1.9 Registry UserAssist (NTUSER.DAT)

|                |                                                                 |
|----------------|-----------------------------------------------------------------|
| File / Inode   | Users/Paulo/NTUSER.DAT — inode 92635                            |
| Method         | TSK icat -o 104448   strings   grep -i qbittorrent              |
| Reason for TSK | Autopsy Registry module did not complete (ingest interruptions) |

Strings recovered:

{6D809377-6AF0-444B-8957-A3773F02200E}\qBittorrent\qbittorrent.exe  
C:\Users\Paulo\Downloads\qbittorrent\_5.2.2\_x64\_setup.exe  
C:\Program Files\qBittorrent\qbittorrent.exe  
qBittorrent.File.Torrent\_.torrent

GUID {6D809377-...} is the well-known Program Files folder identifier, confirming qBittorrent was launched from the standard installation path. The ProgID qBittorrent.File.Torrent\_.torrent confirms the user opened .torrent files through the Windows shell.

## 8.2. Network Analysis (Phase 05)

### 8.2.1 Capture Overview

|                              |                         |
|------------------------------|-------------------------|
| Total packets                | 585,916                 |
| Primary filter (port 29823)  | 25,027 packets (4.3%)   |
| Download window filter       | 20,628 packets (3.5%)   |
| BitTorrent protocol (window) | 276,744 packets (47.2%) |
| HTTP packets (window)        | 6                       |
| TCP peer connections         | 64                      |

### 8.2.2 DHT Traffic

The primary filter revealed extensive BT-DHT traffic. DHT (Distributed Hash Table) is a decentralized peer-discovery mechanism that does not rely on a central tracker. Within the 37-second download window, all 20,628 DHT packets carried the same info-hash, 58846860f0a766f8a42b0bb214d8c713fdf1b167 — identical to the .torrent filename recovered from disk. The concentration of DHT queries for a single info-hash within the exact download window is direct network corroboration of the disk evidence.

**Note.** Filtering for tcp.port == 29823 returned zero results within the download window. This is expected: port 29823 is the inbound listening port. Outbound connections used ephemeral ports (>49152) assigned by the OS. The download was driven by outbound connections.

### 8.2.3 Tracker HTTP Communication

Three HTTP GET /announce requests were sent to 130.239.18.158:6969 (standard BitTorrent tracker port) within the download window.

|           |                                                         |
|-----------|---------------------------------------------------------|
| Tracker   | 130.239.18.158:6969                                     |
| Request   | GET /announce (event=started)                           |
| info_hash | 58846860f0a766f8a42b0bb214d8c713fdf1b167                |
| peer_id   | -qB5220- (qBittorrent 5.2.20)                           |
| port      | 29823 (matches disk artifacts)                          |
| left      | 791,674,880 bytes (matches .torrent bencoding)          |
| Response  | HTTP/1.1 200 OK — Server: mimosa — peers600 (100 peers) |
| Frames    | 14610 (request) / 14680 (response)                      |

The peer\_id -qB5220- independently identifies the client as qBittorrent 5.2.2 — a fourth independent source corroborating the software identity established by Prefetch, application log, and installation artifacts in Phase 04.

**Clarification.** Tracker vs. web seed: the application log recorded a 404 from cdimage.debian.org at 12:29:35. This originated from an HTTP web seed URL declared in the torrent metadata, not from the tracker. The tracker at 130.239.18.158:6969 returned 200 OK. Tracker and web seed are distinct mechanisms.

### 8.2.4 BitTorrent Handshake — Info-Hash in Payload

The first BitTorrent handshake was observed at frame 16107. The handshake has a fixed 68-byte structure:

```
[1 byte] 0x13 protocol name length (19)
[19 bytes] 'BitTorrent protocol'
[8 bytes] reserved extension flags
[20 bytes] info-hash
[20 bytes] peer ID
```

In the hex dump of frame 16107, 'BitTorrent protocol' is visible at offset 0x30 and the info-hash 58846860f0a766f8a42b0bb214d8c713fdf1b167 appears at offsets 0x50–0x60, transmitted in clear text. This is the definitive disk-to-network correlation: the same identifier found on disk appears verbatim in a captured network packet.

### 8.2.5 TCP Peer Connections

Wireshark Conversations revealed 64 TCP connections established from 10.0.2.15 during the download window, all using ephemeral source ports. The complete list is in peers.csv.

| Peer IP               | Est. Volume Received | Note                |
|-----------------------|----------------------|---------------------|
| 152.173.104.78        | ~321 MB              | Largest contributor |
| 82.124.84.162         | ~39 MB               | High-volume seeder  |
| 82.64.35.92           | ~39 MB               | High-volume seeder  |
| 85.216.19.245         | ~38 MB               | High-volume seeder  |
| + 60 additional peers | Variable             | See peers.csv       |

A search for peer IPs in the application log returned no results — expected, as qBittorrent does not log individual peer IPs. Disk-to-network correlation is established through four convergent data points (Section 9).

### 8.2.6 I/O Graph — Traffic Burst

The Wireshark I/O graph with the download-window filter shows near-zero traffic until approximately 120 seconds, a sharp burst peaking near 35,000 packets/second between ~120–150 seconds, and an immediate return to near-zero at ~150 seconds. This burst aligns with the 37-second download window established from disk artifacts, providing independent graphical confirmation of the download event.

## 9. Disk-to-Network Correlation

Four independent data points converge between disk artifacts (Phase 04) and network evidence (Phase 05), establishing the evidentiary link without reliance on individual peer-IP logging.

### 1 Info-hash identity

Disk: .torrent filename = 58846860f0a766f8a42b0bb214d8c713fdf1b167.torrent (inode 156645). Network: same value in the BitTorrent handshake payload (frame 16107). The info-hash is a SHA-1 digest; collision is computationally infeasible.

### 2 Temporal window

Disk: log records torrent added 12:29:33, complete 12:30:10; NTFS \$SI = 12:29:33. Network: DHT queries for the target info-hash begin at 12:29:33; the I/O burst aligns with the 37-second window.

### 3 Port declaration

Disk: qBittorrent.ini Session\Port = 29823; log 'Listening on TCP/29823, UDP/29823'. Network: tracker announce parameter port=29823.

### 4 File size

Disk: .torrent bencoding length = 791,674,880 bytes. Network: tracker announce parameter left=791,674,880.

Cross-reference:

| Evidence Point   | Disk Source (Ph.04)             | Network Source (Ph.05)                |
|------------------|---------------------------------|---------------------------------------|
| Torrent identity | .torrent filename = 58846860... | Info-hash in handshake (16107)        |
| Download start   | log: 12:29:33                   | DHT + announce event=started          |
| Download end     | log: 12:30:10                   | I/O burst ends; piece exchange ceases |
| P2P port         | qBittorrent.ini: 29823          | announce: port=29823                  |
| File size        | bencoding: 791674880            | announce: left=791674880              |
| Client identity  | Prefetch / log PID 3716/8060    | peer_id -qB5220- (5.2.2)              |
| Peer discovery   | log: DHT ENABLED                | 20,628 DHT packets, target hash       |
| Tracker contact  | log: 404 web seed (not tracker) | HTTP 200 OK from tracker              |

## 10. Conclusions

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Based on the totality of evidence across disk analysis (Phase 04) and network analysis (Phase 05), the following conclusions are established with high confidence:

**1. qBittorrent 5.2.2 was installed on the target system.**

Confirmed by browser download history (sourceforge.net, 2026-06-19 10:29 BRT), Prefetch (first run 10:33:04, count 3), installer Prefetch, application log (PID 3716), and Registry UserAssist.

**2. debian-13.5.0-amd64-netinst.iso was downloaded via BitTorrent.**

Confirmed by the application log (added 12:29:33, complete 12:30:10), NTFS timestamps (\$SI=\$FN = 12:29:33 BRT), browser history (cdimage.debian.org, 12:29 BRT), and .torrent bencoding.

**3. The user manually initiated the download.**

Confirmed by Recent Documents .lnk files for both the .iso and .torrent files, proving direct interaction through the Windows shell.

**4. Network traffic corroborates disk evidence independently.**

The info-hash 58846860f0a766f8a42b0bb214d8c713fdf1b167 was recovered from disk and independently identified in BitTorrent handshake payloads. Four convergent data points link disk and network evidence.

**5. No artifact tampering was detected.**

NTFS analysis confirmed \$SI = \$FN on the .torrent file, ruling out timestomping. The timeline is internally consistent across all sources.

**6. The download completed successfully in 37 seconds.**

Despite a web seed 404 error, the client completed the 755 MiB download via 64 peer connections, consistent with DHT-based peer discovery and normal swarm behaviour.

## 11. Limitations

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**No hardware write blocker.** The investigation used a virtualized environment without a physical write blocker. Integrity was maintained through hash verification and careful handling, but physical write blocking was not feasible.

**Initial acquisition invalidated.** The first acquisition was invalidated due to VirtualBox snapshot architecture. Consolidation modified the base VDI; the original snapshot chain is preserved on two backup devices.

**Autopsy ingest incomplete.** Ingest was interrupted twice, preventing the Registry module from completing. Registry artifacts were extracted via TSK as an alternative.

**Network analysis on evidence machine.** Phase 05 was performed on the evidence-generating machine. Findings are corroborated by disk evidence through four convergence points, mitigating this deviation.

**Plaso timeline incomplete.** Autopsy's Plaso timeline module errored. The timeline was reconstructed manually from individual artifact timestamps.

**22.2% packet drop.** Wireshark reported ~22.2% packet loss during capture. Sufficient evidence was recovered for all objectives, but the capture is not a complete record.

**Controlled lab environment.** Evidence was generated by the examiner in a single-investigator lab. Findings demonstrate methodological competence, not analysis of independently sourced evidence.

## 12. Recommendations

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- 1. Use hardware write blockers.** Connect all evidence media through a certified hardware write blocker before acquisition to eliminate the risk of writes to the source.
  - 2. Image snapshot differentials directly.** When acquiring from VirtualBox VMs, point qemu-img at the most recent snapshot differential VDI rather than the base VDI; the differential chain resolves automatically without modifying files.
  - 3. Configure the investigator workstation before analysis.** Install and configure all analysis tools, including Wireshark, on the investigator VM before evidence collection to prevent procedural deviations.
  - 4. Hash all transfers between systems.** Accompany any evidence transfer with SHA-256 verification before and after to ensure integrity.
  - 5. Configure hash sets in Autopsy.** Configuring NSRL known-good and notable hash sets would enable automatic identification of known files, reducing manual effort.
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## 13. Examiner Declaration

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I, **Paulo Vaz**, hereby declare that:

1. This report accurately reflects the findings of the digital forensic examination conducted on the evidence described herein.
2. The examination was performed using recognized forensic tools and methodologies aligned with ISO/IEC 27037, 27041, 27042, 27043 and NIST SP 800-86.
3. All procedural deviations from standard forensic practice have been documented, justified, and disclosed within this report and the chain of custody.
4. The findings, conclusions, and opinions expressed are my own and reflect an honest assessment of the evidence examined.
5. This report was prepared for academic portfolio purposes and does not constitute a legal instrument in any jurisdiction.

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**Paulo Vaz**

Digital Forensic Examiner

Date: \_\_\_\_\_ / \_\_\_\_\_ / 2026

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*Generated June 28, 2026. Evidence image MD5: 8e01029edeadbfff36fe8516afb54df. Case reference: BTD-2026-001.*